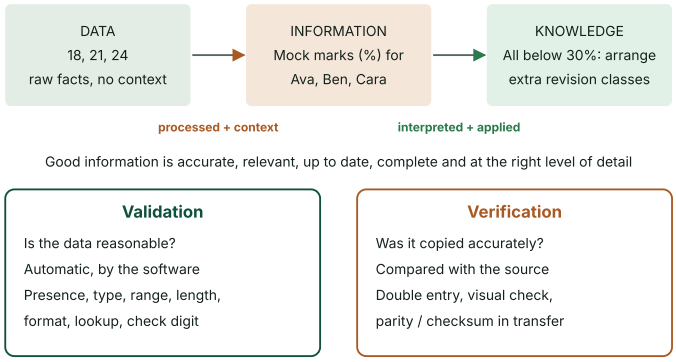
Data, information and knowledge; quality of information; validation and verification

FD06 FACT FILE

Specification: students should be able to

- Distinguish between data, information and knowledge by using examples
- Describe how the quality of information is affected by accuracy, relevance, being up to date, completeness, presentation and reliability
- Describe and apply the range, type, length, format, presence and lookup validation checks
- Explain the purpose of the verification methods double entry and proofreading
- Explain the limitations of data validation and data verification

Key words: data, information, knowledge, context, GIGO, transcription error, transposition error, accuracy, relevance, currency (up to date), completeness, reliability, validation, range check, format check, lookup check, verification, double entry



Data becomes information when it is processed and given context, and information becomes knowledge when it is interpreted and applied; validation and verification compared

Data, information and knowledge

Data is a collection of raw facts and figures – numbers, text, codes, dates, images or sounds – that has no context and therefore no meaning on its own. The sequence 11.2, 9.6, 12.4 could be anything. When data is processed and given context, for example by labelling it as the bleep-test levels of three under-16 rugby players, it becomes information, which has meaning for the person using it. This is often summarised as information = data + context + meaning. Knowledge goes one step further: it is produced by applying a rule, or a set of rules, to information so that an informed decision can be made. If the coach's rule is that any player below level 10 needs extra conditioning, then knowing that one particular player must join that group is knowledge. The distinction matters because organisations collect data in order to produce information, and they use information, together with their own rules and experience, to make decisions. In an answer, always show all three levels using the same example so that the progression from raw values, to meaningful information, to a decision based on a rule is clear.

Quality of information and GIGO

The quality of information depends on the quality of the data used to produce it, an idea summed up as 'garbage in, garbage out' (GIGO). Errors creep in in several ways. A transcription error happens when data is mistyped, such as 1500 entered instead of 150. A transposition error happens when two characters swap places, such as 40719 becoming 47019. Spreadsheets may contain incorrect formulae, which can be caught by calculating important totals in two different ways and comparing them. Surveys may produce biased or meaningless results if questions are poorly worded or only one group of people is asked. Good-quality information has six characteristics: it is accurate, relevant to the user and purpose, up to date, complete enough for the decision, well presented in a suitable format, and reliable because it comes from a trustworthy, verified source. Using poor information can be worse than having none, because people act on it with false confidence and may make costly or dangerous decisions.

Value and cost of information

Information is valuable to an organisation when it leads to better decisions, lower costs, better customer service or an advantage over competitors. A supermarket that knows which products sell on which days can order the right stock and reduce waste; a hospital that knows its bed occupancy can plan staff rotas. The value of information falls if it arrives too late to influence a decision, if it is inaccurate or incomplete, or if competitors have the same information.

Information also has a cost: collecting the data, entering and checking it, storing it securely, keeping it up to date and training staff to use it all cost money. An organisation should therefore only collect information whose value is greater than its cost. Where information is about identifiable people, it is personal data, and it must be processed lawfully, fairly and securely under the UK GDPR and the Data Protection Act 2018, which also requires that data is accurate, kept up to date and not held for longer than necessary.

Validation checks

Validation is an automatic check made by software when data is entered, to confirm that the input obeys set rules and is reasonable and sensible. A range check makes sure a value lies between a lower and an upper limit, such as a shoe size from 1 to 15. A type check makes sure the data is of the correct type, such as an integer for a number of tickets. A length check makes sure the right number of characters has been entered, such as exactly 11 characters for a mobile number. A format check makes sure the data matches a pattern, such as two letters followed by four digits. A presence check makes sure a field has not been left blank, which is why online forms mark compulsory fields with an asterisk. A lookup check compares the entry with a list or file of valid values, such as checking that a membership number exists in the members file. Often several checks are needed on one field, because a format check alone would accept an impossible date such as 45/33/87.

Verification methods

Verification checks that data has been entered or copied accurately, so that every copy exactly matches the original source. In double entry, the data is entered twice and the computer compares the two versions, flagging any difference; this is why websites ask you to type a new password or email address twice. Double entry catches most typing mistakes but doubles the workload, so for large volumes it is usually kept for the most important fields. In proofreading, or visual checking, the data on screen or on a printout is compared by eye with the source document. It works best when a different person does the checking, as in a pharmacy where a second pharmacist checks the printed label against the prescription. Proofreading is slow and expensive for large amounts of data and errors can be overlooked when people rush. Many systems also display the entered details and ask the user to confirm them, and backup software verifies that the copy matches the original.

Limitations of validation and verification

Validation and verification reduce errors but cannot remove them completely. Validation only checks that data is reasonable, so a value that is sensible but wrong will be accepted: an age of 34 typed instead of 43 passes any sensible range check. Some impossible values can also slip through if the rules are too simple; a day-of-month check of 1 to 31 accepts 31 April.

Verification only checks that the copy matches the source. If the original paper form contains a mistake, or the customer gave the wrong information, then the stored data will be an exact copy of a wrong value. Double entry can be defeated when the person makes the same mistake twice, and proofreading depends on the concentration of the checker. Because of these limits, organisations combine methods: validation rules at the point of entry, verification of the most important fields, and processes for people to check and correct their own records, which is also a right under data protection law.

T Key terms from the mark schemes

Data
Raw facts and figures, such as numbers, text, images or sound, that have no context and therefore no meaning on their own.
`Credited: Raw facts and figures without context`

Information
Data that has been processed and given context so that it has meaning to the person using it.
`Credited: Data + context + meaning`

Knowledge
The result of applying one or more rules to information so that an informed decision can be made.
`Credited: Information that has been interpreted and applied using experience or rules`

GIGO
Garbage in, garbage out: if the input data is of poor quality, the output information will also be of poor quality.

Transposition error
An input error in which two characters are entered in the wrong order, such as 47019 instead of 40719.

Transcription error
An input error in which data is copied or typed incorrectly, such as 1500 instead of 150.

Validation
An automatic check carried out by software to make sure that input data is sensible and reasonable and obeys set rules.
Credited: Checks data is within acceptable boundaries

Range check
A validation check that a value lies between a lower and an upper limit.

Format check
A validation check that data matches a set pattern of characters, such as two letters followed by four digits.
Credited: Picture check

Lookup check
A validation check that compares the entered value with a list or file of valid values.
Credited: List check

Verification
A check that data has been entered or copied accurately, so that it exactly matches the original source.

Double entry
A verification method in which data is entered twice and the computer compares the two versions, flagging any difference.

C C# example

Validation in C#

```
string[] forms = { "11A", "11B", "11C" };
Console.Write("Form class: ");
string form = Console.ReadLine() ?? "";
if (form == "") Console.WriteLine("Presence check failed");
else if (Array.IndexOf(forms, form) == -1) Console.WriteLine("Lookup check failed");
else Console.WriteLine("Form class accepted");

Console.Write("Year group (8-14): ");
string input = Console.ReadLine() ?? "";
if (input == "") Console.WriteLine("Presence check failed");
else if (!int.TryParse(input, out int year)) Console.WriteLine("Type check failed");
else if (year < 8 || year > 14) Console.WriteLine("Range check failed");
else Console.WriteLine("Year group accepted");
```

Exam focus

Questions on this lesson usually use State, Describe, Explain or Compare and carry one mark per valid point. For data, information and knowledge, mark schemes credit a definition plus an example for each, ideally taken from the table or scenario given: raw value, value with context, then a rule applied to reach a decision. Knowledge answers that do not mention applying a rule usually lose the mark. For quality factors, name the factor and explain its effect in the context, for example that an incomplete re-order message does not say which pack size to buy. For validation, name the check and give the precise rule with the field and limits, such as 'range check: hour >= 0 and <= 23'. 'Checks the data is correct' is never credited for validation; say 'sensible' or 'reasonable'. For verification, describe how the method works and a drawback. Common mistakes are confusing a type check with a format check, describing double entry as validation, and claiming that either method guarantees accuracy.

? Check yourself

1. A hospital ward records patients' temperatures. Using an example from this ward, show how data, information and knowledge differ. [3]
2. Explain what is meant by 'garbage in, garbage out' (GIGO). [2]
3. A pharmacy technician enters a patient number of 40719 as 47019. State the type of error that has been made and explain why validation may not detect it. [3]

Answers: 1. Data: a raw value with no context, e.g. 38.9 (1) / Information: the data with context/meaning, e.g. patient in bed 4 had a temperature of 38.9 °C at 14:00 (1) / Knowledge: a rule applied to the information to make a decision, e.g. rule 'above 38 °C means fever', so the nurse alerts the doctor (1) 2. If the input data is of poor quality / inaccurate (1) / the output information will also be poor / inaccurate, however good the processing system is (1) 3. Transposition error (1) / 47019 is still a sensible / reasonable patient number, e.g. correct length, type and format (1) / validation only checks data is reasonable, not that it is correct (1). Credit: a check digit could detect it.